

W8 – The Natural Logarithm

MHF4U

Jensen

1) Use a calculator to approximate each to the nearest thousandth

a) $\ln 6.2$

$$\approx 1.825$$

b) $\ln 2.1$

$$\approx 0.742$$

c) $\ln e$

$$= 1$$

d) e^5

$$\approx 148.413$$

2) Expand each logarithm

a) $\ln x^2$

$$= 2 \ln x$$

b) $\ln \sqrt[3]{x}$

$$= \ln(x^{1/3})$$

$$= \frac{1}{3} \ln x$$

c) $\ln \frac{u^3}{wv^4}$

$$= \ln(u^3) - \ln(wv^4)$$

$$= 3 \ln(u) - [\ln(w) + \ln(v^4)]$$

$$= 3 \ln(u) - \ln(w) - 4 \ln(v)$$

3) Condense each expression to a single logarithm

a) $4 \ln 2$

$$= \ln(2^4)$$

$$= \ln(16)$$

b) $\ln 10 - 5 \ln 7$

$$= \ln(10) - \ln(7^5)$$

$$= \ln\left(\frac{10}{7^5}\right)$$

$$= \ln\left(\frac{10}{16807}\right)$$

c) $3 \ln x + 3 \ln y$

$$= \ln(x^3) + \ln(y^3)$$

$$= \ln(x^3 y^3)$$

4) Solve each equation. Round your answer to 4 decimal places if necessary.

a) $e^x = 2$

$$x = \ln(2)$$

$$x \approx 0.6931$$

b) $e^{-3n} = 83$

$$-3n = \ln(83)$$

$$n = \frac{\ln(83)}{-3}$$

$$n \approx -1.4729$$

c) $e^{k+7} = 26$

$$k+7 = \ln(26)$$

$$k = \ln(26) - 7$$

$$k \approx -3.7419$$

d) $9e^{1.4p-10} - 10 = 17$

$$9e^{1.4p-10} = 27$$

$$e^{1.4p-10} = 3$$

$$1.4p-10 = \ln(3)$$

$$p = \frac{\ln(3)+10}{1.4}$$

$$p \approx 7.9276$$

$$\text{e) } \ln x = -5$$

$$e^{-5} = x$$

$$x \approx 0.0067$$

Rule: $b^{\log_b(x)} = x$

$$\text{f) } 7.316 = e^{\ln(2x)}$$

$$7.316 = 2x$$

$$x = \frac{7.316}{2}$$

$$x \approx 3.658$$

Alternate Method

$$\ln(2x) = \ln(7.316)$$

$$2x = 7.316$$

$$x = \frac{7.316}{2}$$

$$x \approx 3.658$$

$$\text{g) } \ln(-m) = \ln(m + 10)$$

$$-m = m + 10$$

$$-2m = 10$$

$$m = -5$$

$$\text{h) } \ln(9x + 1) = \ln(x^2 + 9)$$

$$9x + 1 = x^2 + 9$$

$$0 = x^2 - 9x + 8$$

$$0 = (x - 8)(x - 1)$$

$$x = 8, 1$$

$$\text{i) } \ln(1 - 8x) - 10 = -7$$

$$\ln(1 - 8x) = 3$$

$$e^3 = 1 - 8x$$

$$8x = 1 - e^3$$

$$x = \frac{1 - e^3}{8}$$

$$x \approx -2.3857$$

$$\text{j) } \ln(5 - 2x^2) + \ln 9 = \ln 43$$

$$\ln(5 - 2x^2) = \ln(43) - \ln(9)$$

$$\ln(5 - 2x^2) = \ln\left(\frac{43}{9}\right)$$

$$5 - 2x^2 = \frac{43}{9}$$

$$5 - \frac{43}{9} = 2x^2$$

$$\frac{2}{9} = 2x^2$$

$$\frac{1}{9} = x^2$$

$$x = \pm \frac{1}{3}$$